Cloud-containerization-and-DevOps

Experiment 2:- Docker Installation, Configuration, and Running Images

Step 1: Pull Image

```
docker --version
```

```
exit
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
```

Step 2: Run Container with Port Mapping

```
docker run -d -p 8080:80 nginx
```

```
Run 'docker run --help' for more information
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker run -d -p 8080:80 --name nginx-container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
58261a837e072ada0319e55846b48ff294b3a02fba6e52bf7c3ec503b5cf2e98
```

Step 3: Verify Running Containers

```
docker ps
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker ps
[sudo] password for palak:
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
d1d16711340f   nginx     "/docker-entrypoint.…"   41 seconds ago   Up 40 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     kind_sutherland
```

Step 4: Stop Container

```
docker stop <container_id>
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker stop d1d
d1d
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                      PORTS     NAMES
d1d16711340f   nginx          "/docker-entrypoint.…"   4 minutes ago   Exited (0) About a minute ago         kind_sutherland
```

Step 5: Remove Container

```
docker rm <container_id>
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker rm d1d
d1d
```

Step 6: Remove Image

```
docker rmi nginx
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Deleted: sha256:4af177a024eb8a1e43f4fb6c66735bb8260115cb5925a64f51673219bd97c144
Deleted: sha256:74821a6b71dfac14eaeb80c7c9e48130391306557f75228278054afe105cde75
Deleted: sha256:ae46c5037a8cc17f658f78b0e6523c9363ca800bf51f91f58cbc3aad8ffe53b5
Deleted: sha256:ec81248cdbc6b02bb24b1c32d4f59194a4a42aab3c81a42631efa1af7cba4f69
Deleted: sha256:774cc13f29089d7d0ebaaa291c5ac2d12841f2d70ab07a21c470d3cbdc11c059
Deleted: sha256:f1155703f3479bf6b295f5e17684156f82f815ca0798af8a2d08cf994411a055
Deleted: sha256:a455653294af11c547d0b3d2d769fd184f357cd5c0decbd1f89ffe9b380985aa
Deleted: sha256:e50a58335e1366e2581fe61794c1651afe2fe04e881e795aa166f24f4fc78d92
```

Result Docker images were successfully pulled, containers executed, and lifecycle commands performed.